

# Wenfán Wu, Ph.D

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## **RESEARCH INTERESTS**

Marine biogeochemical modeling in nearshore zones

Interactions between hydrodynamic and biogeochemical processes

## **WORK**

**Assistant Research Scientist**

Jul 2026 – present

Virginia Institute of Marine Science, W&M

**Postdoctoral Research Associate**

Aug 2024 – Jul 2026

Virginia Institute of Marine Science, W&M

## **EDUCATION**

**Ph.D. in Physical Oceanography**

Sep 2020 – Jun 2024

Ocean University of China

**M.S. in Applied Oceanography** (bypassed to Ph.D. program)

Sep 2018 – Jun 2020

Ocean University of China

**B.S. in Marine Science**

Sep 2014 – Jun 2018

Ocean University of China

## **PROJECTS**

### **2022 – 2027**

Assessing 2035 Climate Change Risk to the Chesapeake TMDL using a Next-generation Unstructured-grid Model (PI: Y. Joseph Zhang), U.S. Environmental Protection Agency

### **2025 – 2026**

Key Technology Development for a Semi-Implicit Cross-Scale Hydrodynamic-Ecological Coupled Forecasting Model (GJHYJBH20241205; PI: Fangguo Zhai), North China Sea Marine Forecasting Center

### **2022 – 2025**

Summer Coastal Upwelling and Its Impact on Bottom Dissolved Oxygen in Marine Ranching Areas off Yantai–Weihai (42176016; PI: Fangguo Zhai), National Natural Science Foundation of China

### **2021 – 2023**

Seasonal-Interannual Variability and Physical Mechanisms of Dissolved Oxygen off Northeastern Shandong Peninsula (ZR2020MD059; PI: Fangguo Zhai), Shandong Province Natural Science Foundation

## **SELECTED PUBLICATIONS**

- 1) Wu, W., Wang, Z., Shen, J., Zhang, Y. J., Testa, J. M., Tian, R., Linker, L. C. (2026) Unveiling drought-induced algal blooms in tidal freshwater zones with an interpretable bloom risk index. *J Geophys Res-Bioge*, (under review).
- 2) Wu, W., Wang, Z., Zhang, Y. J., Shen, J., Tian, R., Linker, L. C., & Cerco, C. F. (2026). Improving cross-scale hydrodynamic simulations in the Chesapeake Bay with physically based calibration. *Ocean Model*, 102680. <https://doi.org/10.1016/j.ocemod.2026.102680>
- 3) Wu, W., Song, C., Chen, Y., Zhai, F., Liu, Z., Liu, C., Gu, Y., & Li, P. (2025). High-frequency dynamics of bottom dissolved oxygen in temperate shelf seas: The joint role of tidal mixing and sediment oxygen demand. *Limnol. Oceanogr.*, <https://doi.org/10.1002/lno.12733>. **(2025 Wiley Top-viewed Article)**
- 4) Wu, W., Zhai, F., Liu, Z., Liu, C., Gu, Y., and Li, P. (2023). The spatial and seasonal variability of nutrient status in the coastal rivers of China shaped by the human activities. *Ecol. Indic.*, 157: 111223. <https://doi.org/10.1016/j.ecolind.2023.111223>.
- 5) Wu, W., Zhai, F., Liu, C., Gu, Y., and Li, P. (2023). Three-dimensional structure of summer circulation in the Bohai Sea and its intraseasonal variability. *Ocean Dynam.*, 1-20. <https://doi.org/10.1007/s10236-023-01576-6>.
- 6) Wu, W., Zhai, F., Gu, Y., Liu, Z., and Li, P. (2023). Weak local upwelling elevates the risks of ecological disasters in shallow waters. *Environ. Re. Lett.*, 18: 114031. <https://doi.org/10.1088/1748-9326/ad0256>.

- 7) Wu, W., Liu, Z., Zhai, F., Li, P., Gu, Y., and Wu, K. (2021). A quantitative method to calibrate the SWAN wave model based on the whitecapping dissipation term. *Appl. Ocean Res.*, 114, 102785. <https://doi.org/10.1088/1748-9326/ad0256>.
- 8) Wu, W., Li, P., Zhai, F., Gu, Y., and Liu, Z. (2020). Evaluation of different wind resources in simulating wave height for the Bohai, Yellow, and East China Seas (BYES) with SWAN model. *Cont. Shelf Res.*, 207, 104217. <https://doi.org/10.1016/j.csr.2020.104217>.

## **CONFERENCES**

### **Jun 2026 - Chesapeake Community Research Symposium (Annapolis, USA)**

Session (Co-lead): *Advancing the Development and Management Applications of Next-generation Airshed, Land-use, Watershed, and Estuarine Models*

### **Nov 2025 - Coastal & Estuarine Research Federation Conference (Richmond, USA)**

Session Talk: *Improving Cross-scale Hydrodynamic Simulation in the Chesapeake Bay with Physically Based Calibration*

### **May 2021 - National Marine Technology Conference (Zhoushan, China)**

Session Talk: *Physical Controls of Summer Variations in Bottom Layer Oxygen Concentrations in the Coastal Hypoxic Region off the North-eastern Shandong Peninsula in the Yellow Sea*

## **ACADEMIC SERVICE**

Peer reviewer for *JGR-Oceans*, *Progress in Oceanography*, *Scientific Reports*, *Ocean Modelling*, and other international peer-reviewed journals.

## **SOFTWARE DEVELOPMENT**

A [MATLAB toolbox](#) designed for the SCHISM model.

A 1-D biogeochemical model established with MATLAB following the CoSiNE frame.

## **PERSONAL WEBSITES**

ResearchGate <https://www.researchgate.net/profile/Wenfan-Wu>

GitHub <https://github.com/wenfanwu>

Personal Homepage <https://wenfanwu.github.io/>